

CS 2490 – Summative Homework 1

Turn in by 2026/9/26 11:59PM

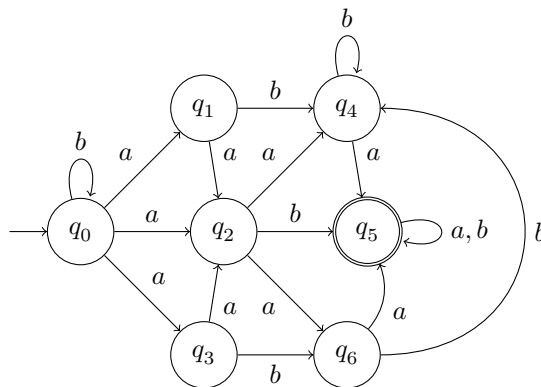
You **must** turn in your assignment as a single zip file entitled: *lastname_firstname_shw1.zip*. It will contain a pdf file *lastname_firstname_shw1.pdf* with your written answers (for which you can use L^AT_EX, etc. or you can write your solutions by hand and scan them), and a JSON file for each relevant exercise named *lastname_firstname_shw1_exX.json* (where X is the exercise number).

Exercise 1. Let Σ be an arbitrary alphabet.

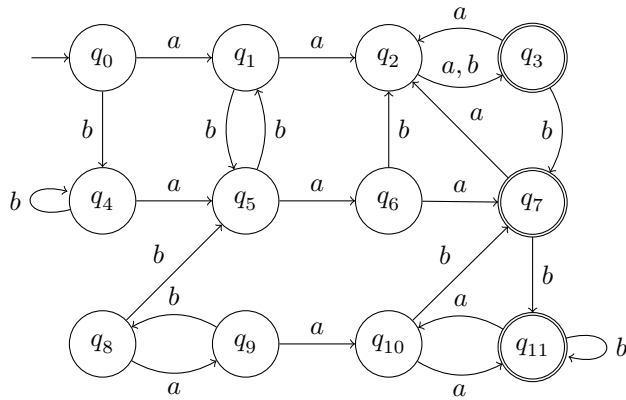
- (i) Show that the empty language $\emptyset$ is regular.
- (ii) Let n be an arbitrary natural number. Suppose you know that every language over Σ with exactly n words is regular. Show that every language over Σ with exactly $n + 1$ words is also regular.¹
- (iii) What does the two previous questions together prove?

¹ Hint: write your language with $n + 1$ words as the union of a language with n words and a language with 1 word.

Exercise 2. Draw the determinization (i.e., the subset construction) of the following NFA. Include the JSON file of the resulting DFA in your submission.



Exercise 3. Draw the minimization of the following DFA. Include in your answer the table/half-table of distinguishability. Include the JSON file of the minimized DFA in your submission.



Exercise 4. Let Σ be an arbitrary alphabet. For a word $w \in \Sigma^*$, we define $\text{even}(w)$ as the word obtained by removing all letters in odd positions of w .

Show that, if a language L over Σ is regular, then the language $\text{even}(L) = \{\text{even}(w) \mid w \in L\}$ is also regular.²

² Hint: suppose you have a DFA that recognizes L , and build from it a NFA that recognizes $\text{even}(L)$.